STM Exercise 3

- Creep
- Fatigue

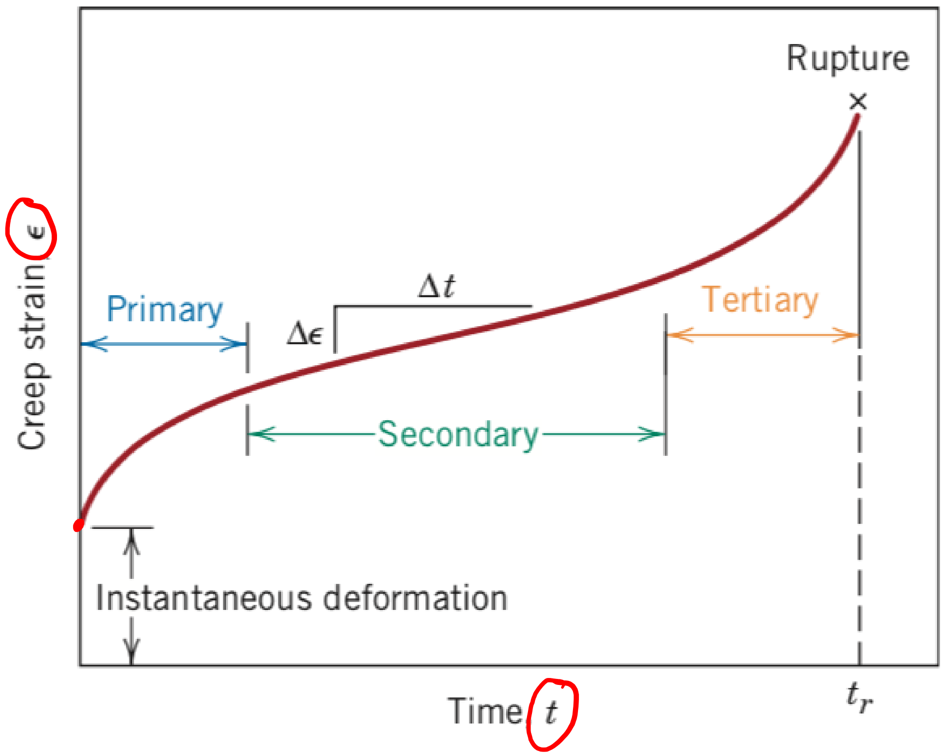
CREEP

creep test: a constant load is applied at a given T, strain is recorded versus time.

Constant load in the elastic field can give to a plastic deformation, progressive to fracture.

1. Thermally activated process (High T Process)

Metals **T>0,3-0,4 T$_{\mathbf{melting}}$**
Ceramics **T> 0,6-0,7 T$_{\mathbf{melting}}$**
Amorphous materials **T>T$_{\mathbf{glass}}$**



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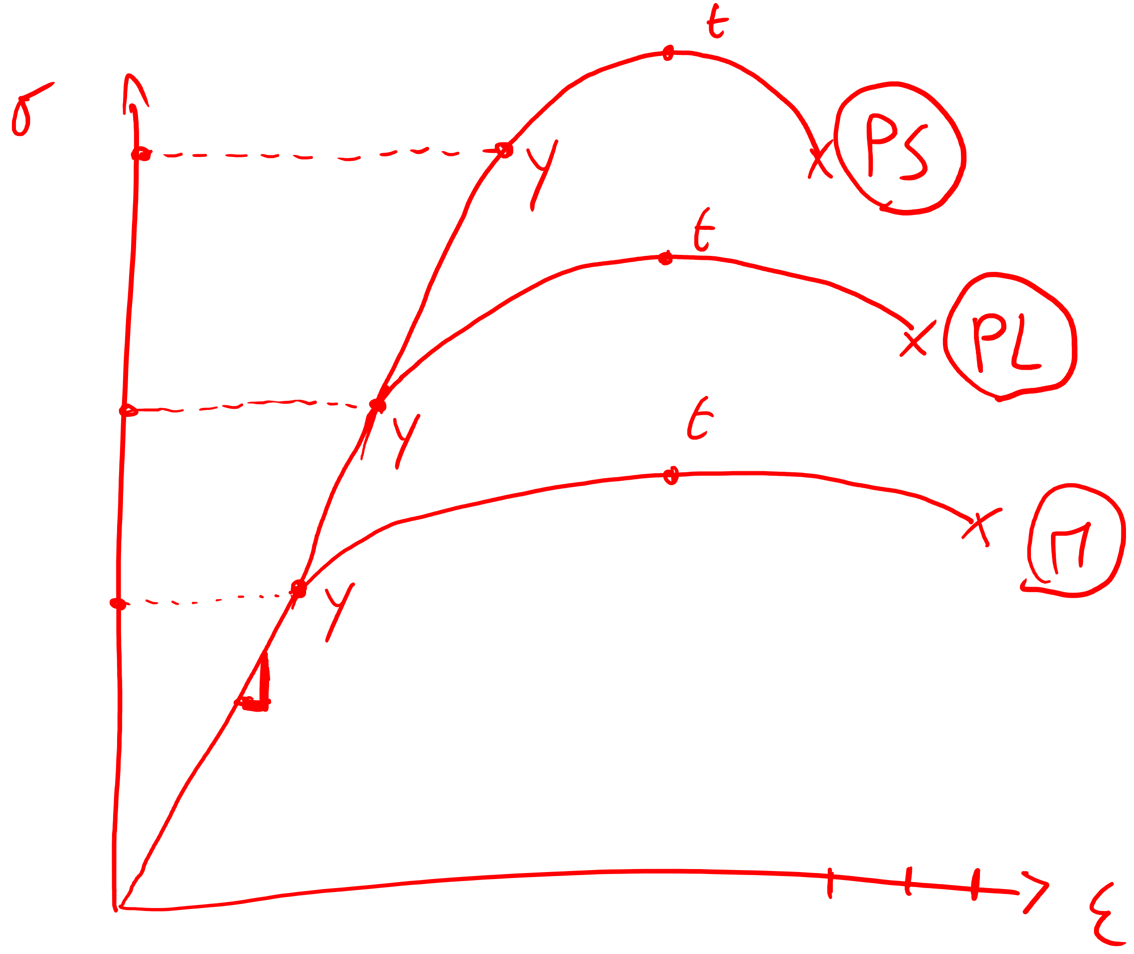
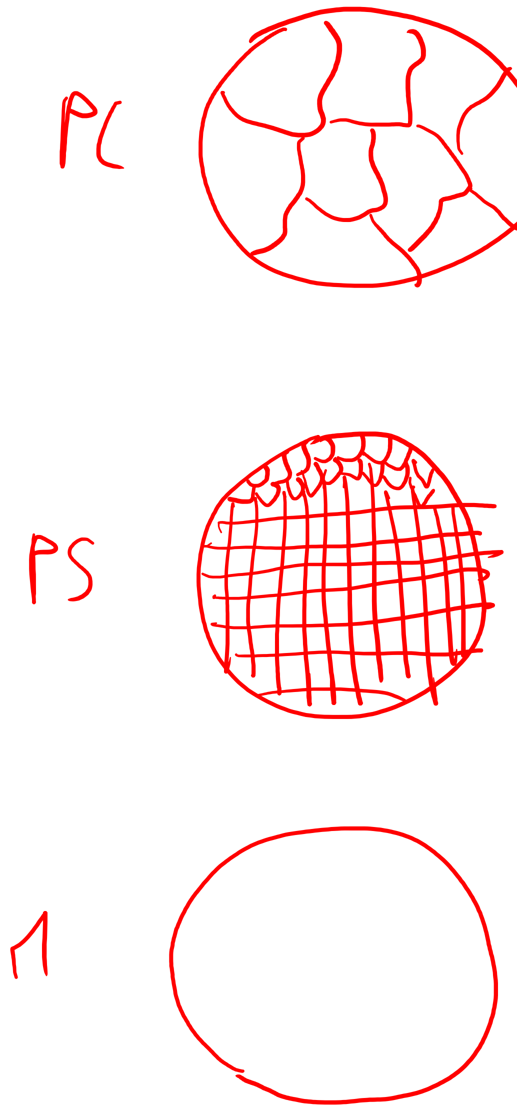
M PL PS





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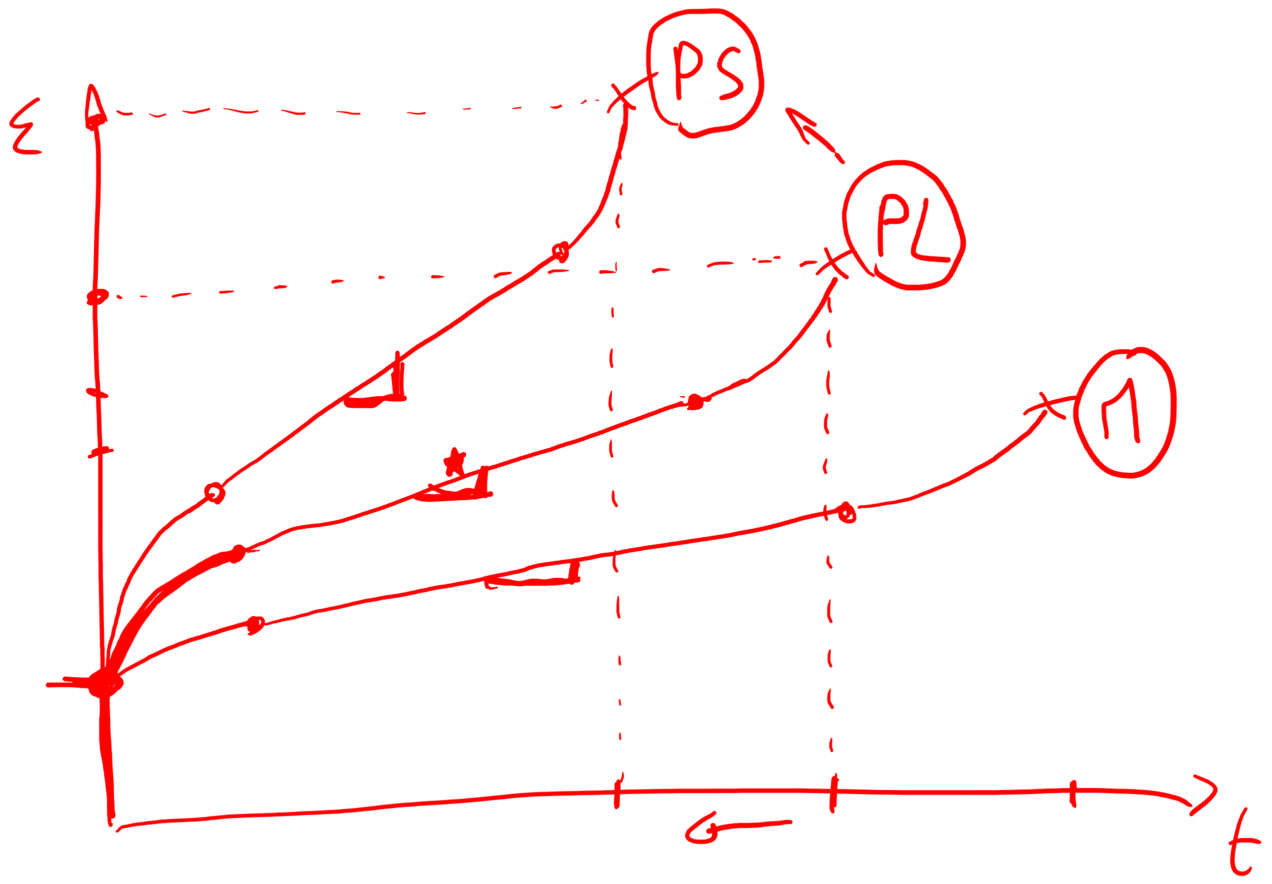
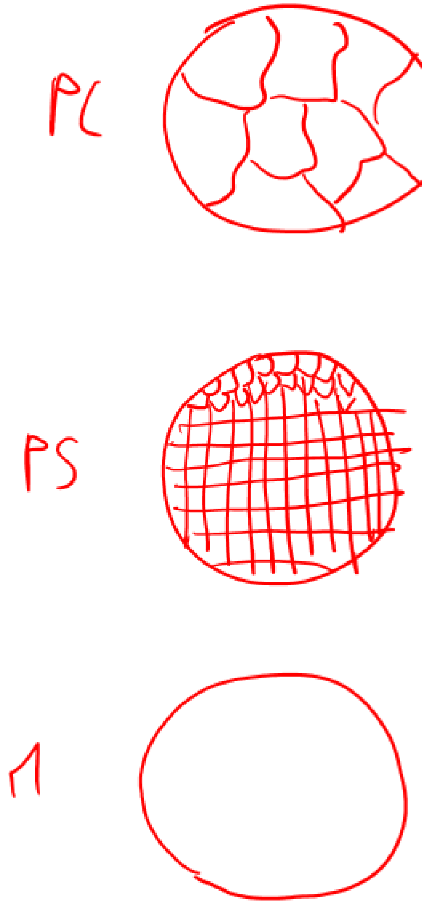
$$\star \ \dot{\varepsilon} = \frac{\Delta \varepsilon}{\Delta t} = \text{Const}$$





2- Draw the creep curve of materials and briefly discuss the three phases of creep in term of materials micro-structural modification. Draw the creep curves of the same material at $\sigma_1 > \sigma_2$ and at T1>T2. Explain differences.

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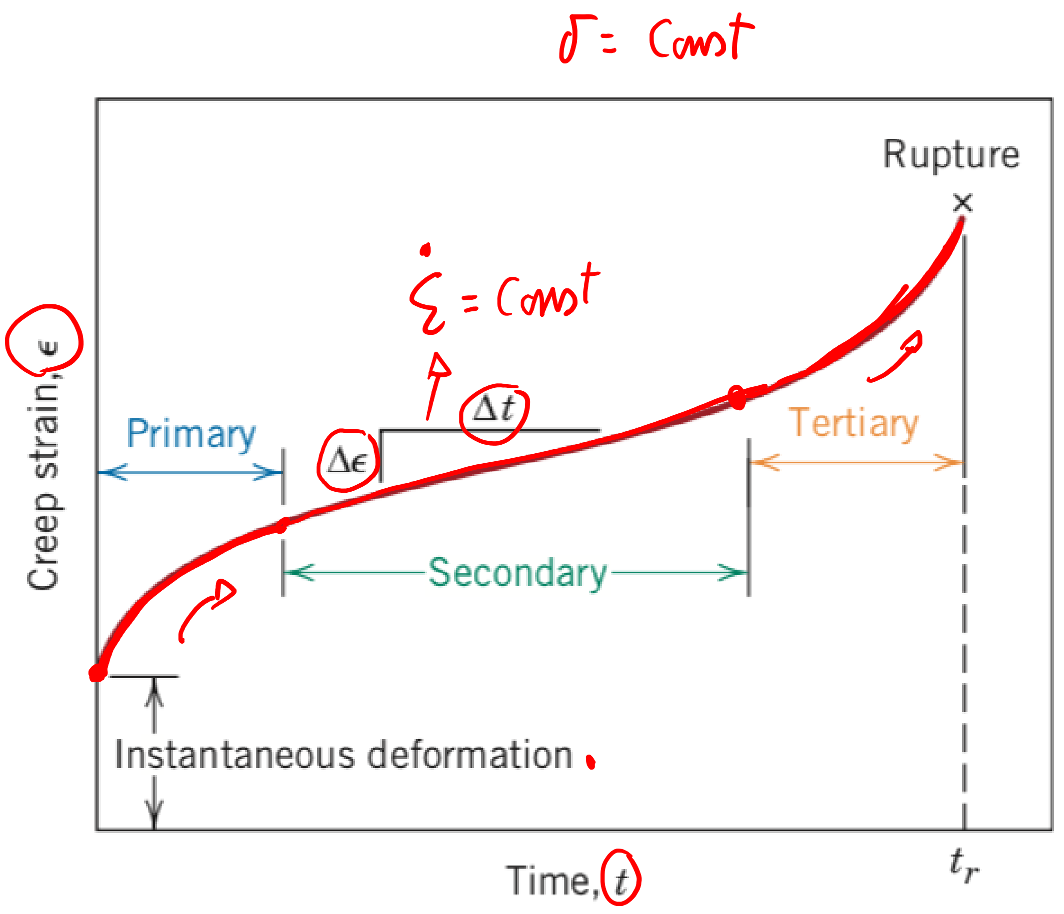

CREEP I:

cold working higher than annealing.

CREEP II:

constant strain: balance between cold working and annealing

CREEP III:

micro-cavities and other macro-defects at the grain boundaries: fracture of the sample

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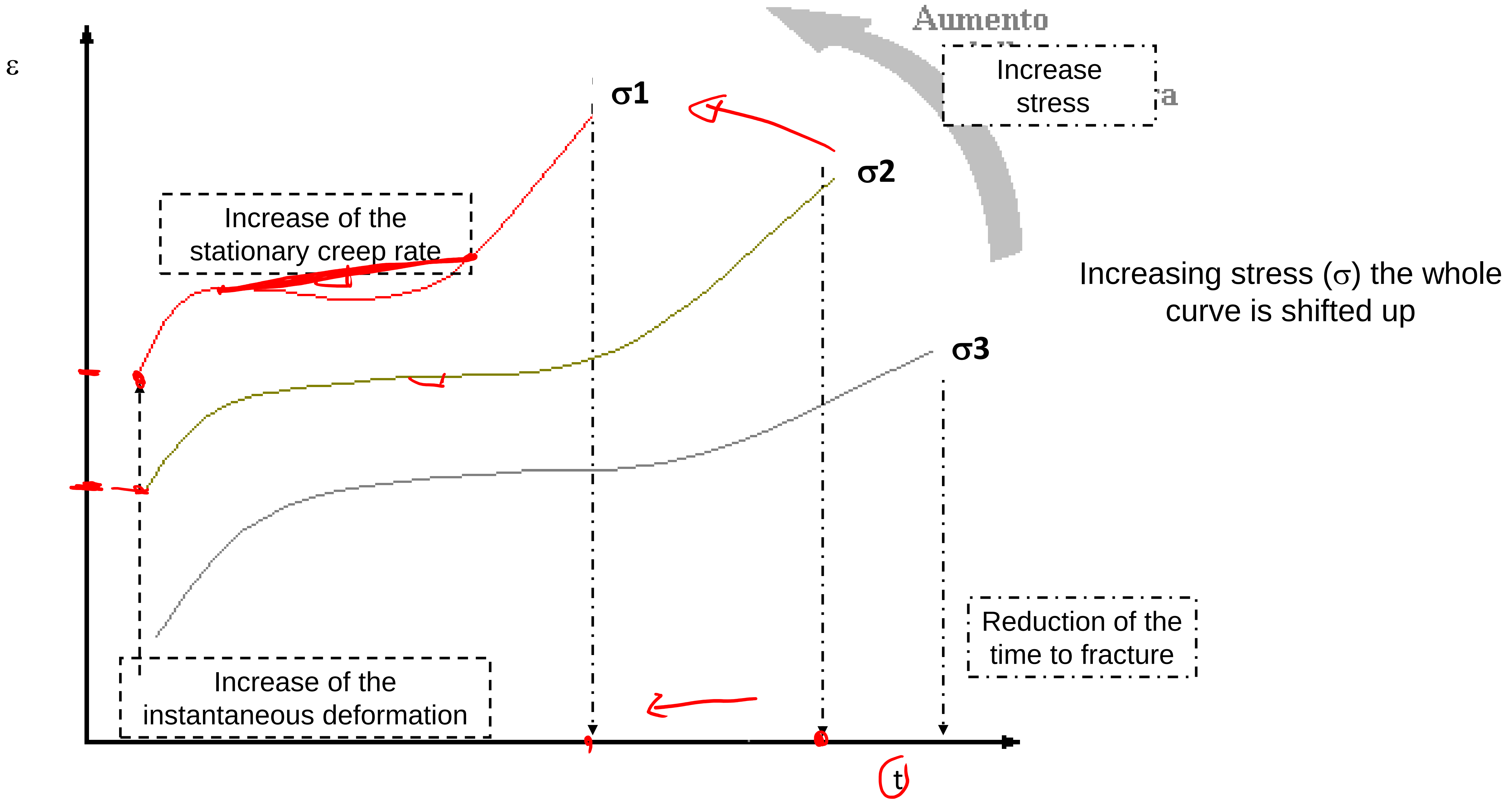

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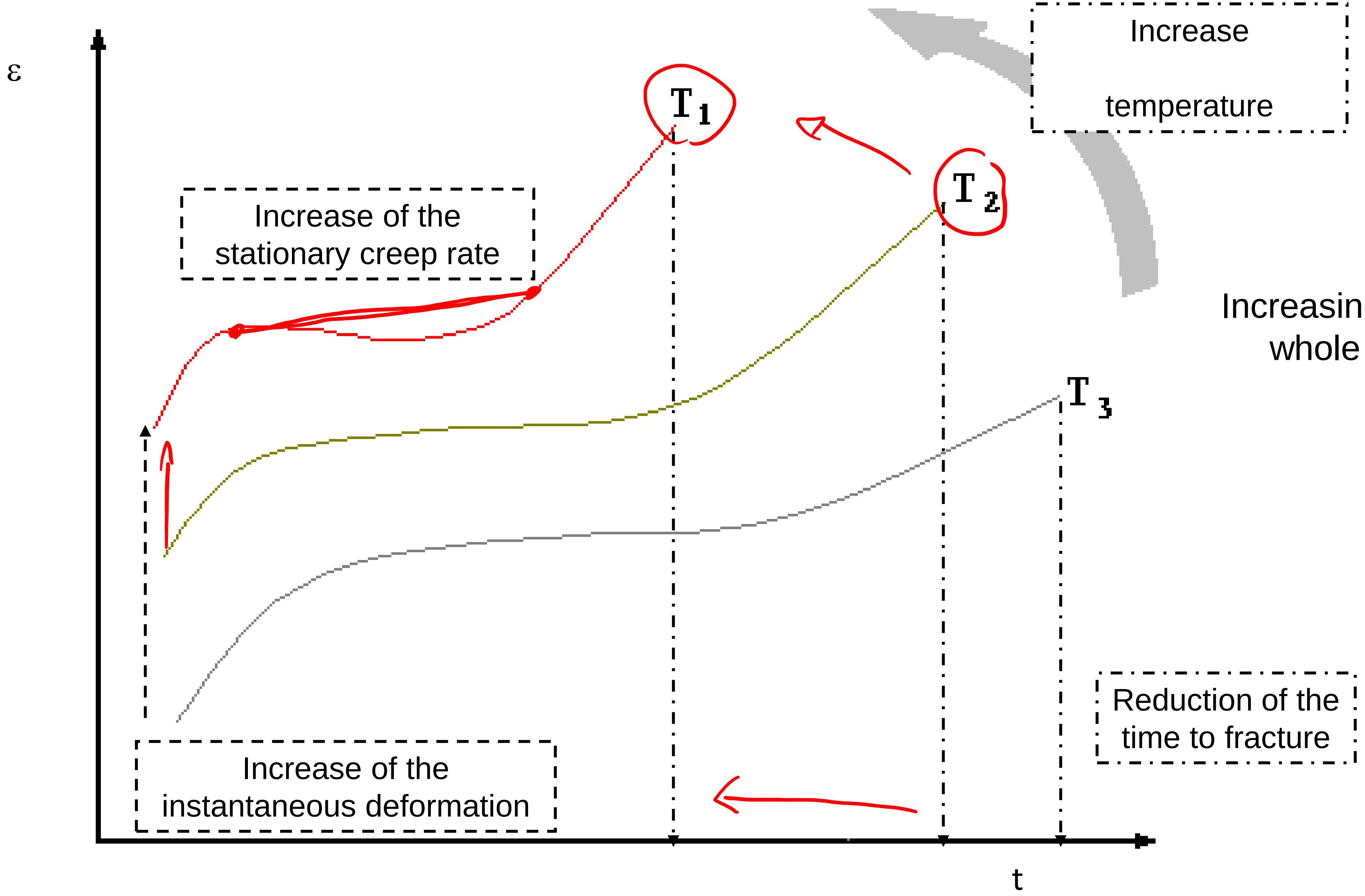
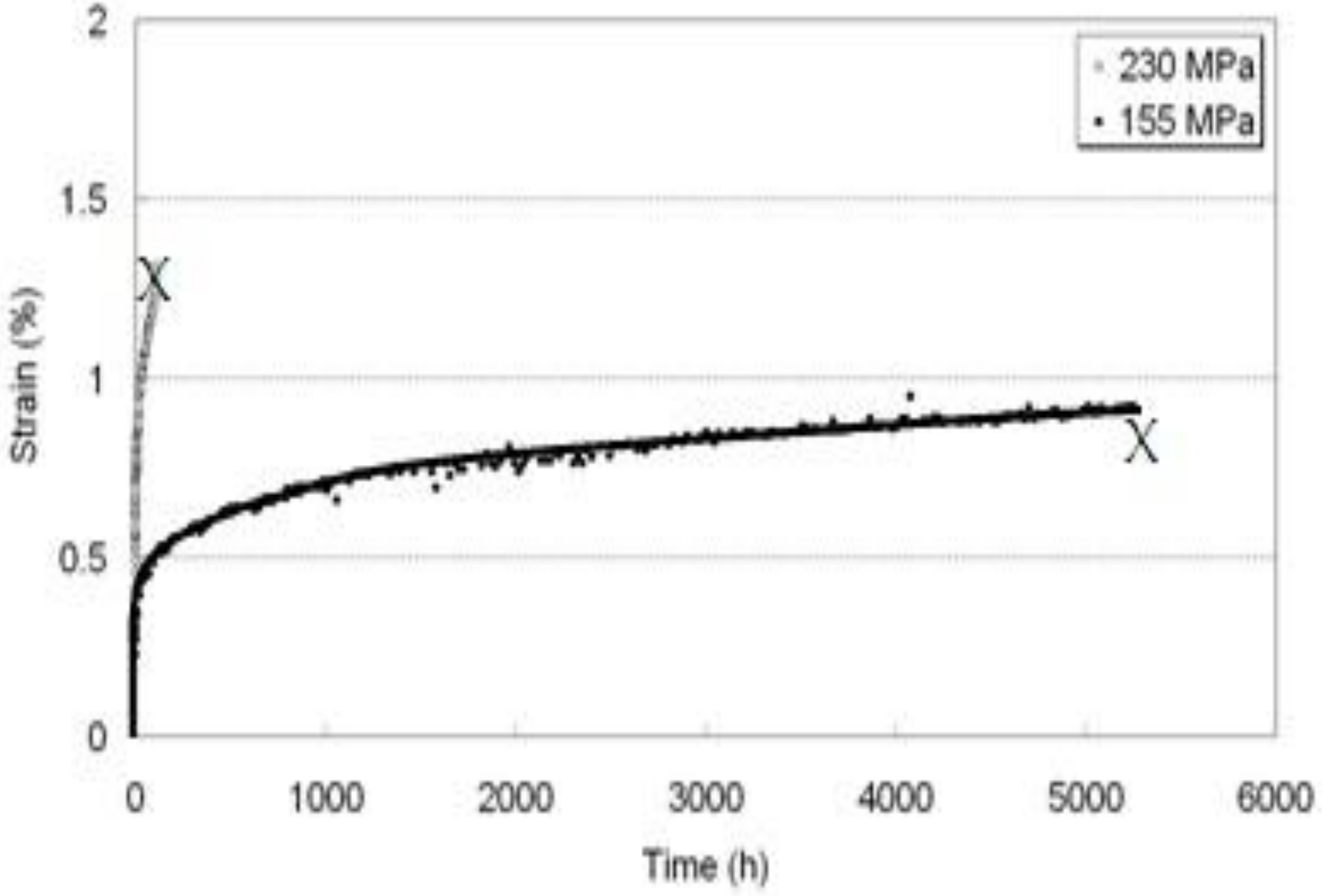


Increasing Temperature (T) the whole curve is shifted up

3- Indicate on the creep curve below the plastic and elastic deformation of a sample loaded at stress=155 MPa after 3000 hours.

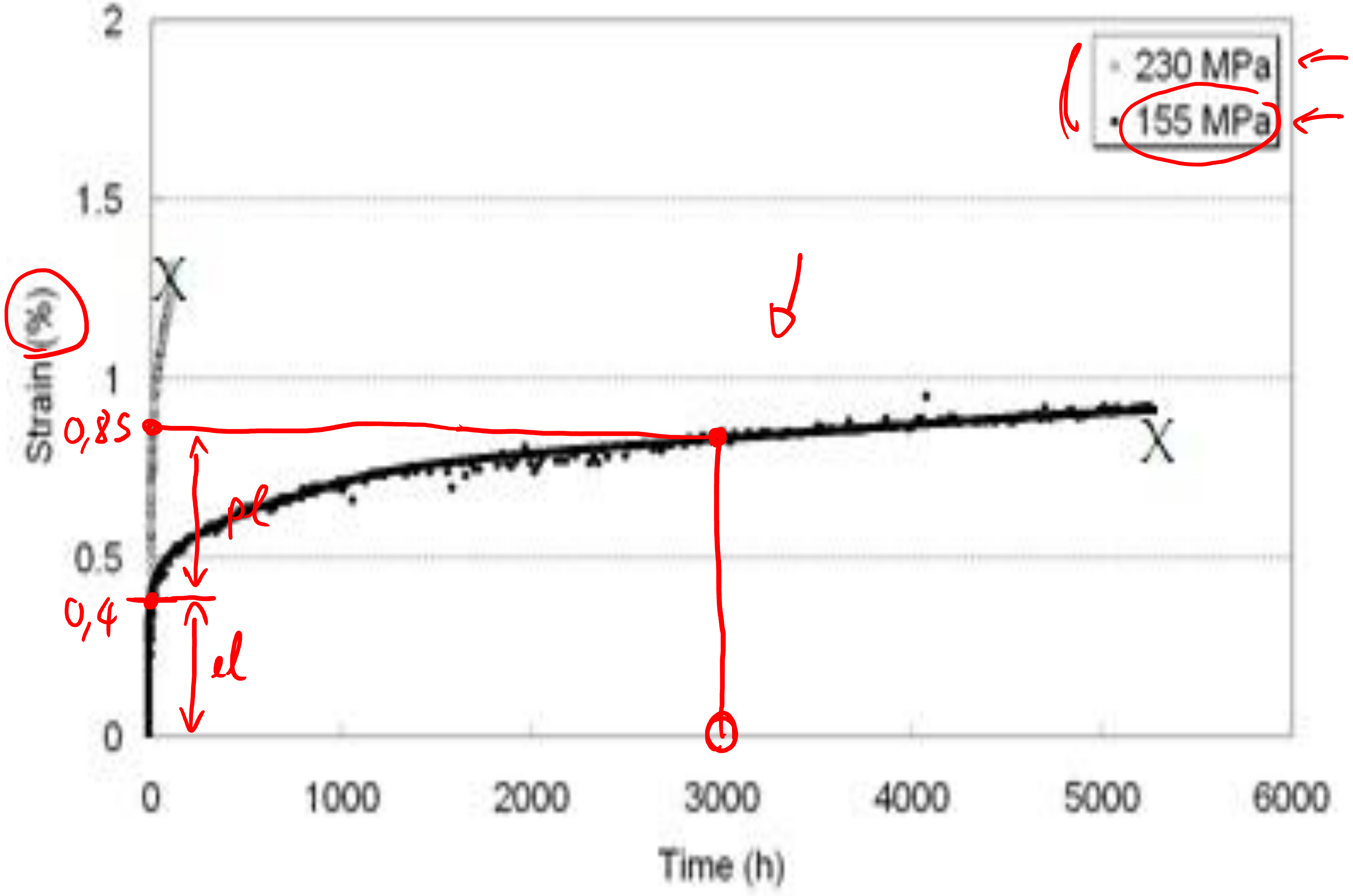


3- Indicate on the creep curve below the plastic and elastic deformation of a sample loaded at stress=155 MPa after 3000 hours.

0,4%

$\sigma =$

$\varepsilon_{pl} = ?$ $\varepsilon_{el} = ?$



$\varepsilon_{el} = 0,4\% = 0,004$

$\varepsilon_{tot} = 0,85\%$

$\varepsilon_{pl} = \varepsilon_{tot} - \varepsilon_{el} =$

$= 0,85 - 0,4 = 0,45\%$

$= 0,0045$

4- Draw the stress/strain curves after tensile tests of a metal at room temperature (RT) and at T1 > RT. Briefly explain the difference.

ε

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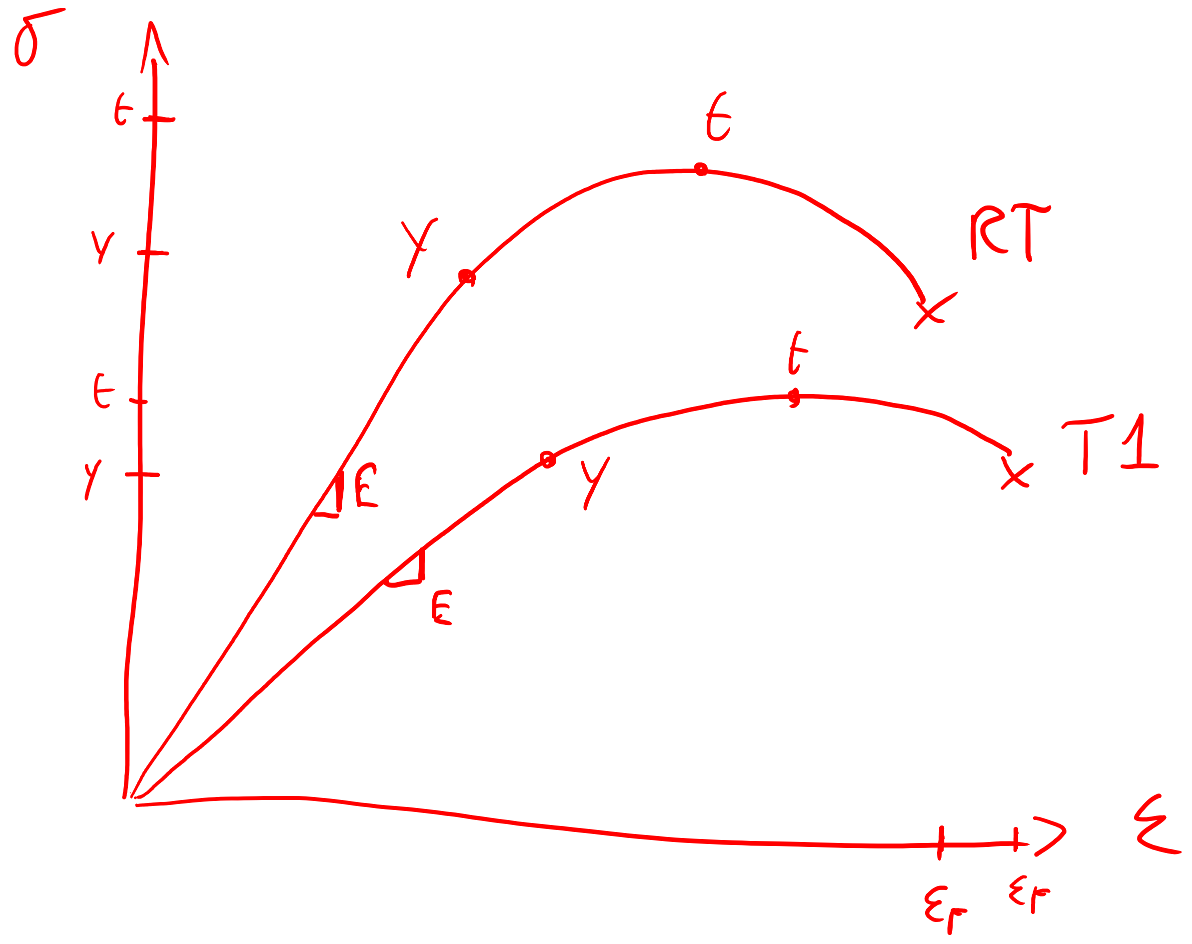
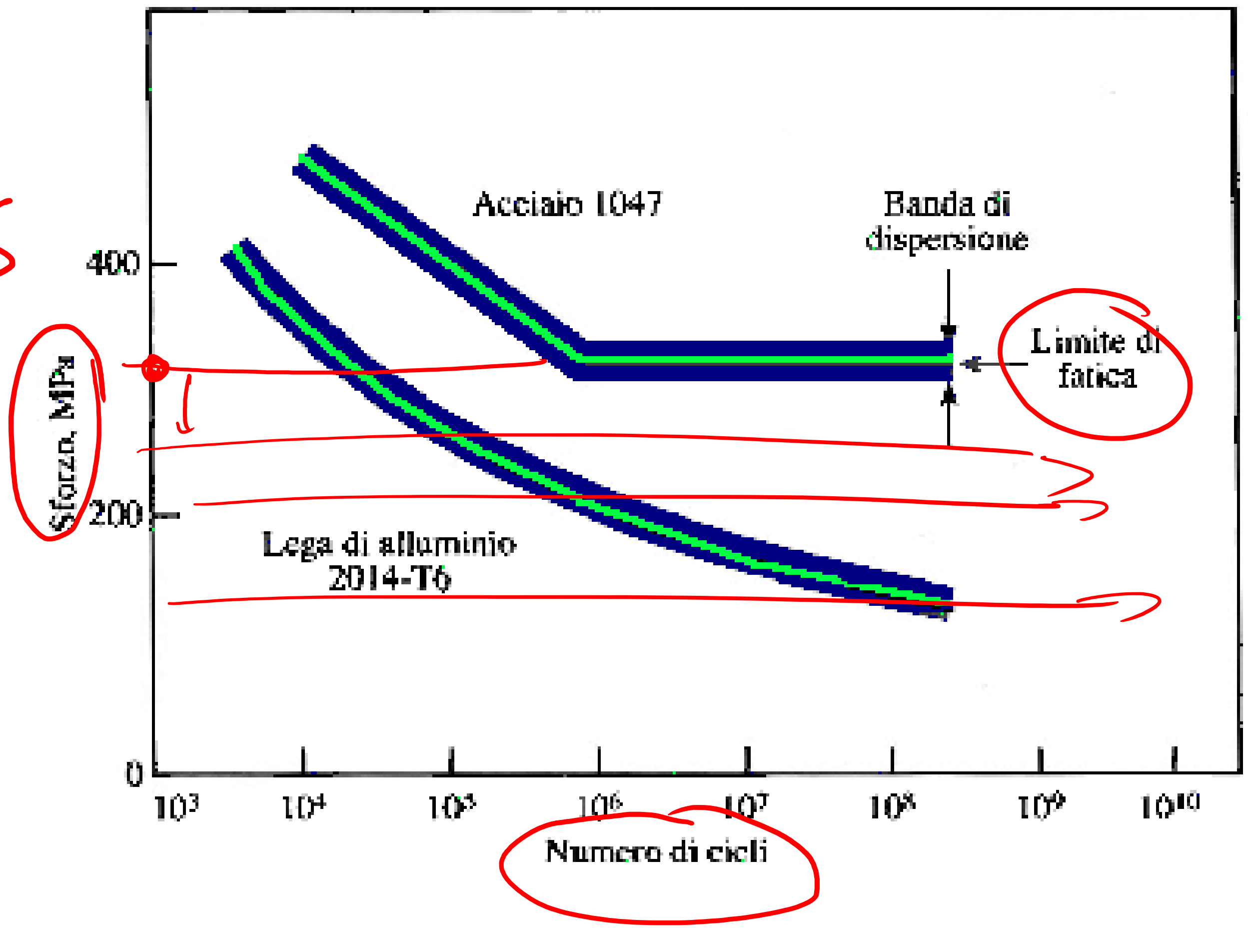


ε

Fatigue

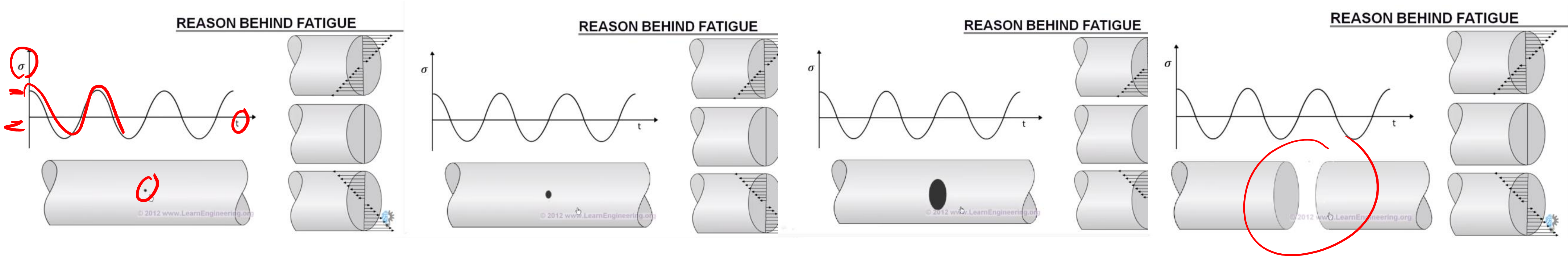
- Failure of materials due to cyclic loading.
- Main reason of mechanical failure of materials
- Failure happens at stress lower than σ_R or σ_Y
- Catastrophic failure of materials (also for ductile materials !)
- Fatigue tests: materials are cyclically loaded at different stresses up to failure.
- Fatigue limit: when cyclically loaded below this limit, materials do not fail (time considered by the graph)

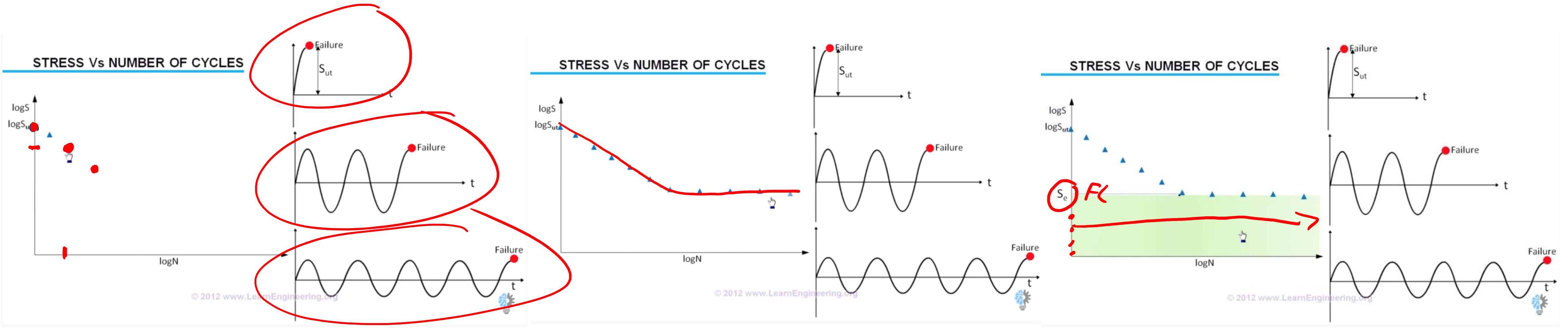
S



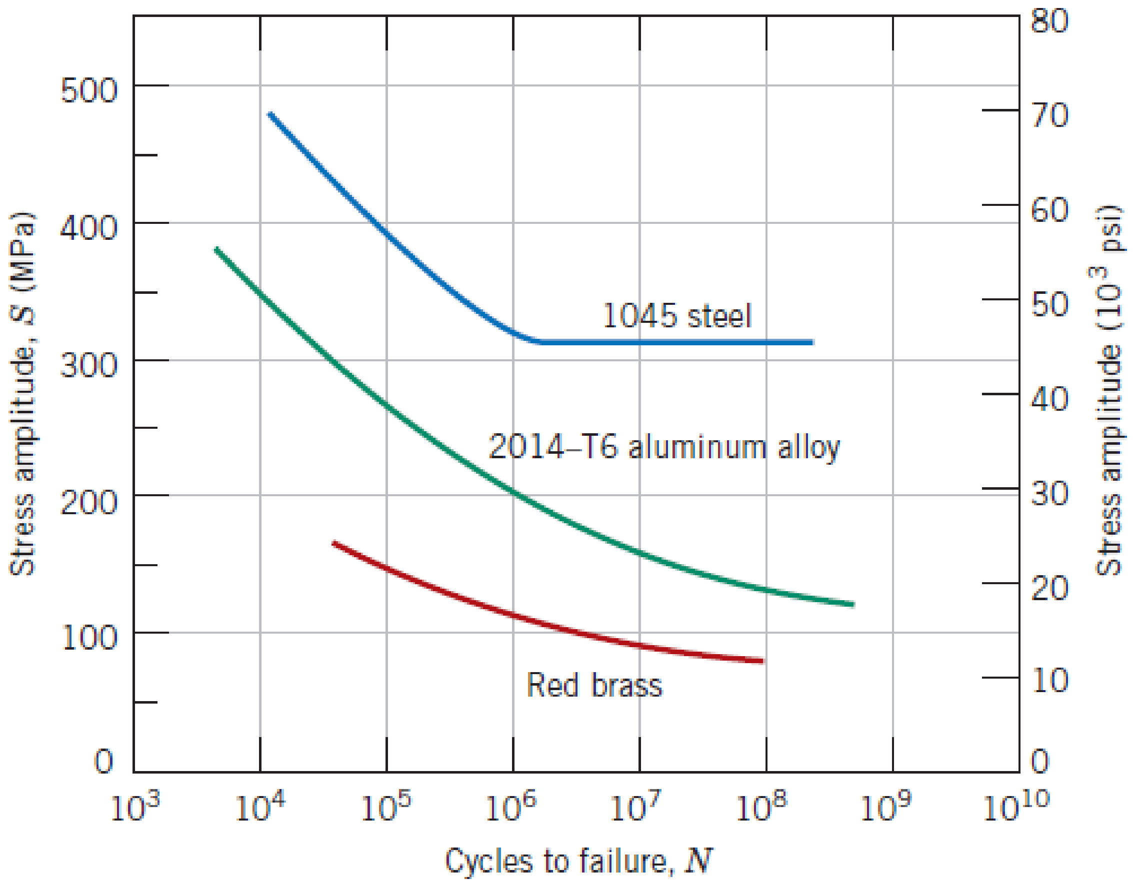
Fatigue

More cycles





5- A Brass cylinder (8 mm diameter) is cyclically loaded at +7500 N e -7500 N along its axis. Use curves below to calculate its fatigue life.



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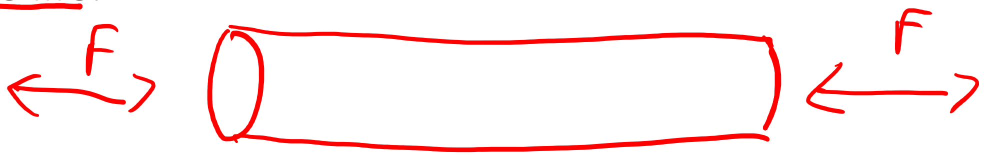
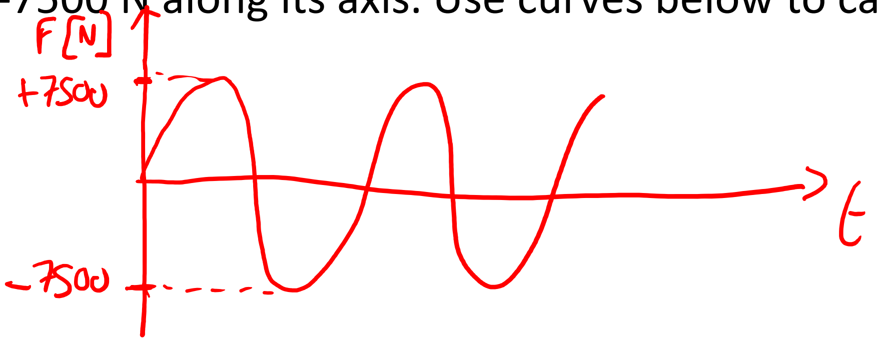
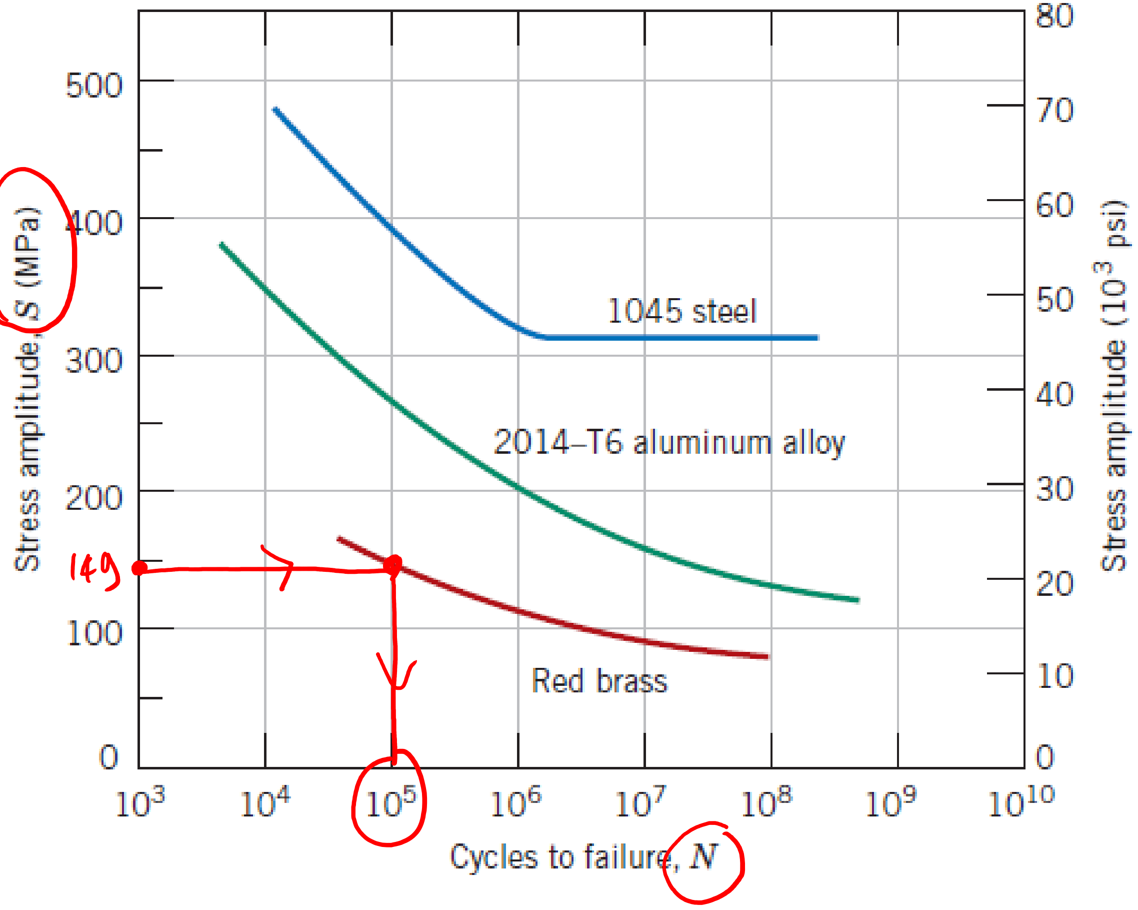




$N = ?$



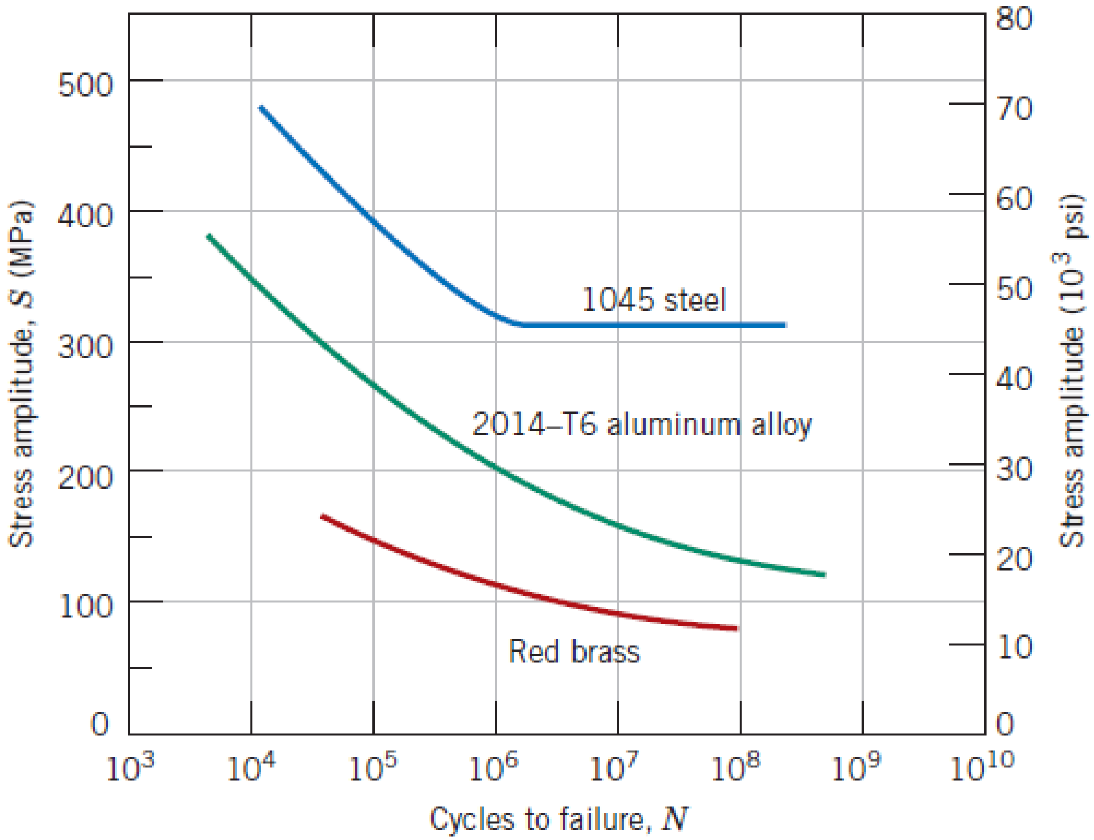
$\sigma = F/A$

$A = \pi \cdot r^2 = 50{,}27\ mm^2$

$\sigma = \dfrac{7500\ N}{50{,}27\ mm^2} = 149{,}2\ MPa$

$N = 10^5$

A steel rod is cyclically loaded at S= 22 kN along its axis. Use curves above to calculate the minimum diameter for this component to avoid fatigue fracture. Use a safety coefficient 2.



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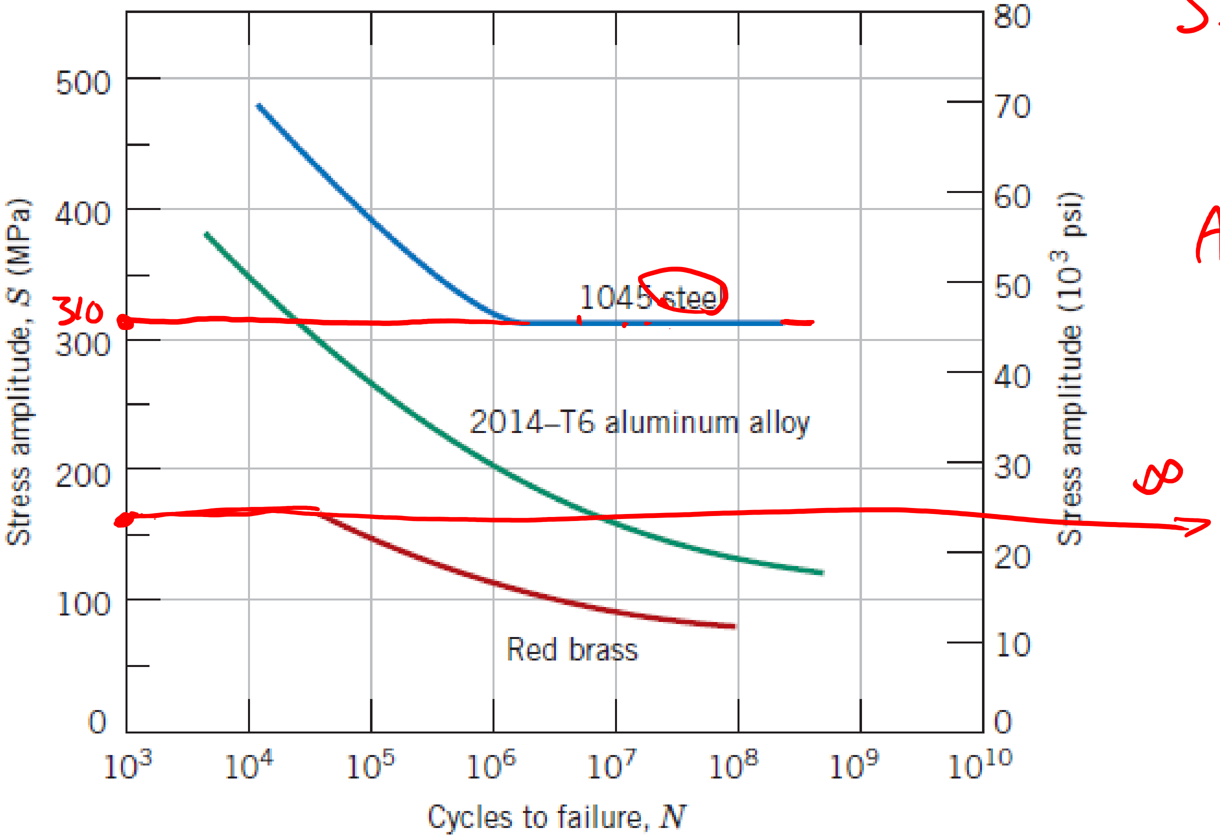
↳ $S < S_{FL}$

→ $\frac{S_{FL}}{2}$

$d = ?$

$S_{FL} = 310\ MPa$



∞

$S = \frac{S_{FL}}{2} = \frac{310}{2} = 155\ MPa$

$S = \frac{F}{A}$

$A = \frac{F}{S} = \frac{22\,000\ N}{155\ MPa} = 141{,}9\ mm^2$

$A = \pi \cdot r^2$

$r = \sqrt{\frac{A}{\pi}} = 6{,}7\ mm$

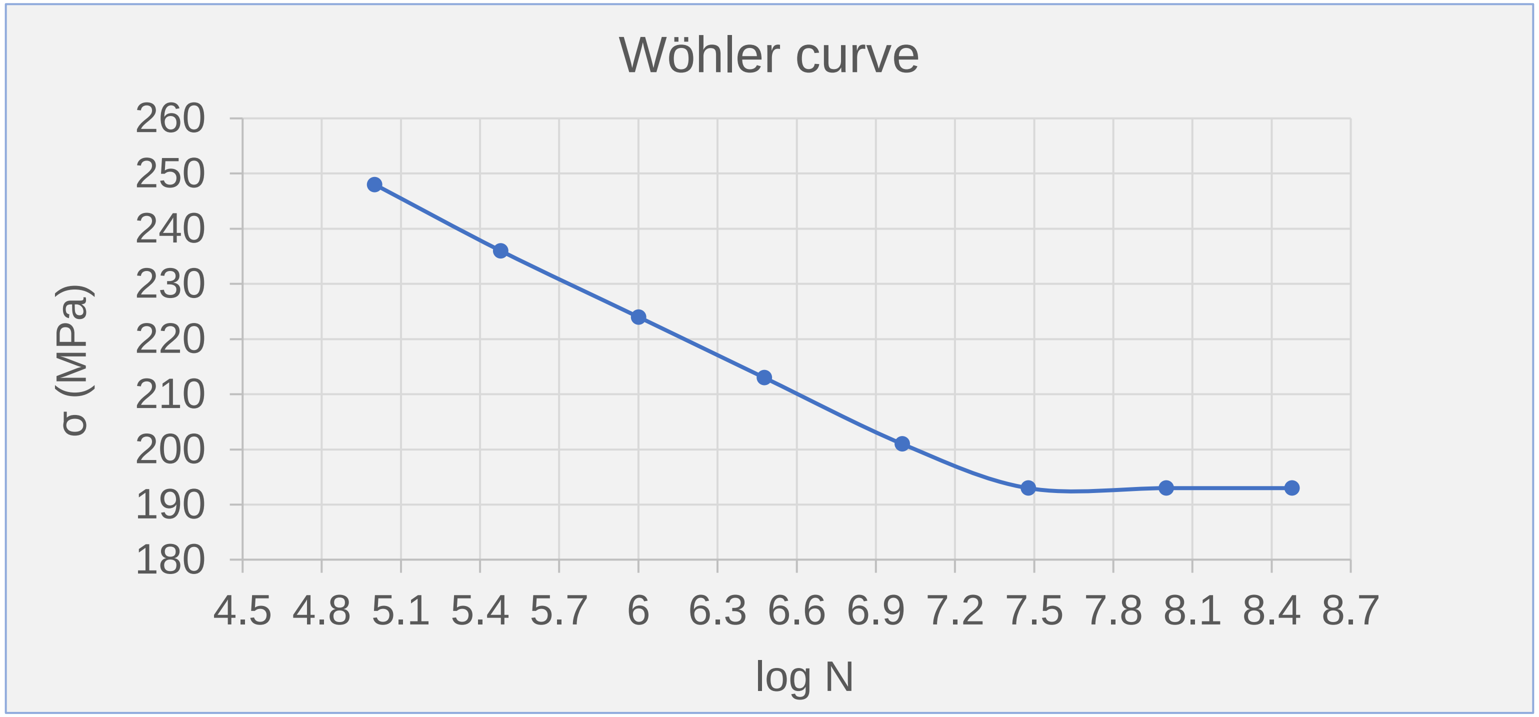
$d = 13{,}4\ mm$

Es 5. The data obtained from a series of fatigue tests for a material are as follows:

σ (MPa)	N°
248	$1 \cdot 10^5$
236	$3 \cdot 10^5$
224	$1 \cdot 10^6$
213	$3 \cdot 10^6$
201	$1 \cdot 10^7$
193	$3 \cdot 10^7$
193	$1 \cdot 10^8$
193	$3 \cdot 10^8$

After constructing the Wöhler curve, determine:

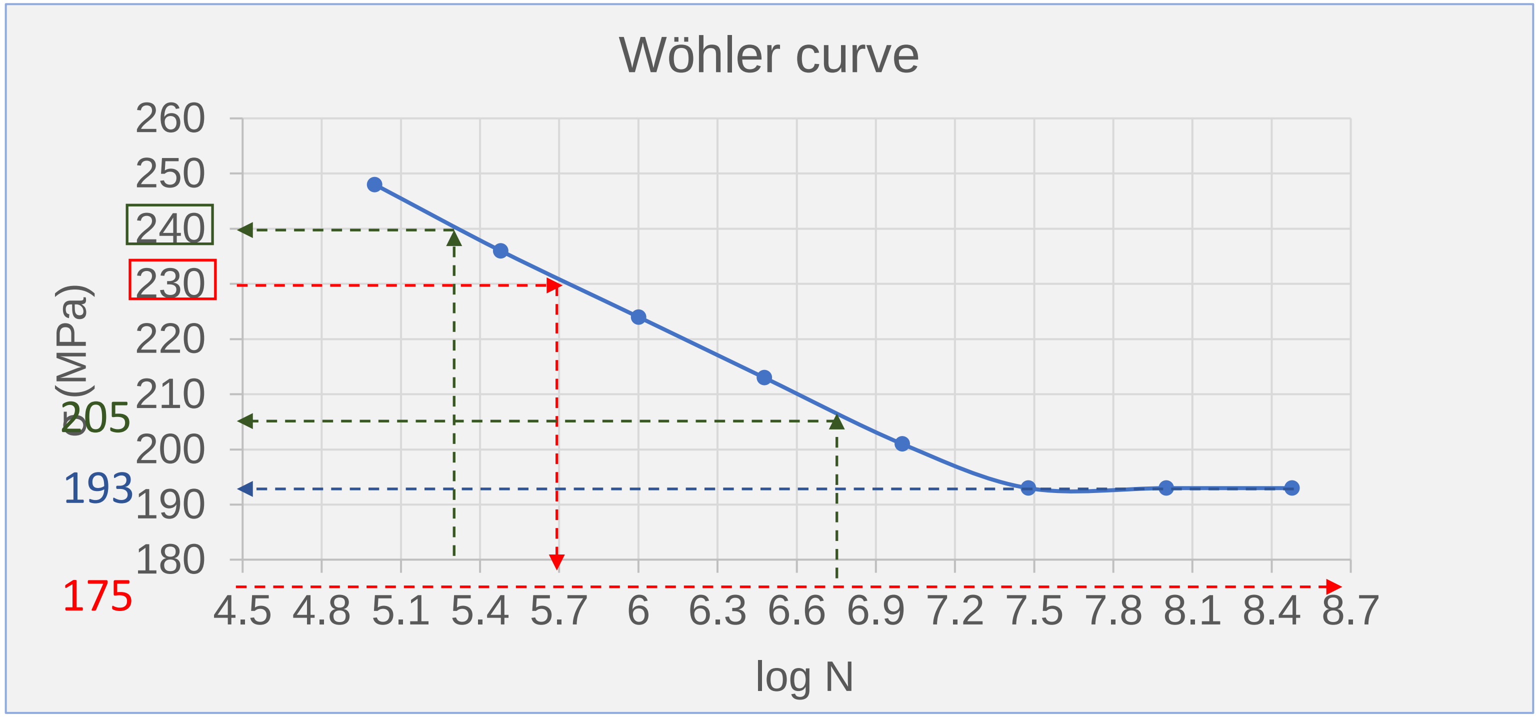
a) The fatigue limit
b) The number of cycles to failure at 230 and 175 MPa
c) The fatigue strength at $2 \cdot 10^5$ and $6 \cdot 10^6$ cycles



a) Fatigue limit :

b) Number of cycles to failure at 230 MPa:
Number of cycles to failure at 175 MPa:

c) Fatigue strength at $2 \cdot 10^5$ cycles:
Fatigue strength at $6 \cdot 10^6$ cycles:



a) Fatigue limit : 193 MPa

b) Number of cycles to failure at 230 MPa: 501 187; N = $5*10^5$

Number of cycles to failure at 175 MPa: ∞

c) Fatigue strength at $2{\cdot}10^5$ cycles: log ($2{\cdot}10^5$) = 5.3 → 240 MPa;
Fatigue strength at $6{\cdot}10^6$ cycles: log ($6{\cdot}10^6$) = 6.8 → 205 MPa

6- Draw the fatigue curve (S/N) of a metal having the following surface characteristics: polished surface; sand-blasted surface; U-notches surface; V-notched surface. Explain differences.

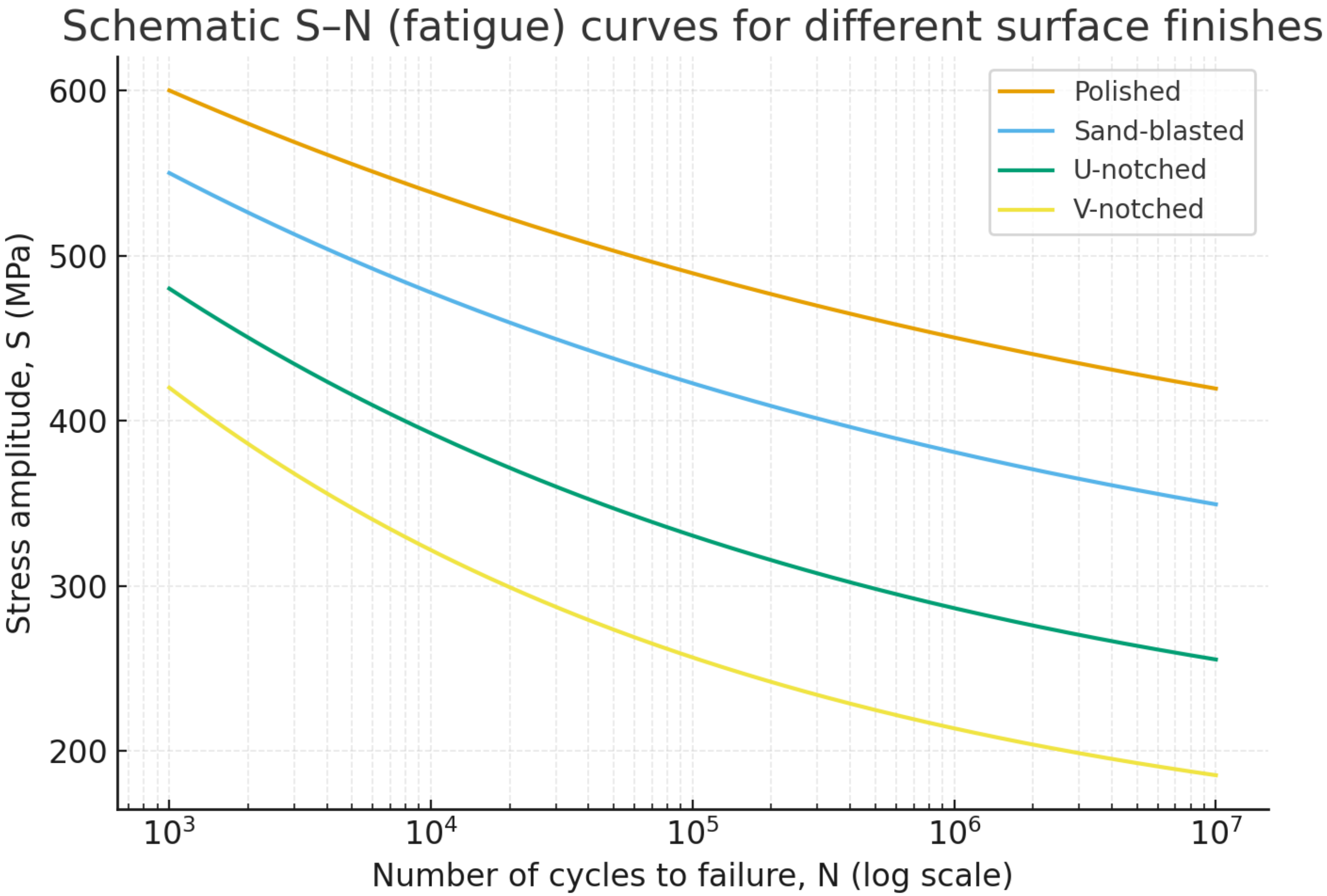
1- Polished

2- Sand-blasted

3- U-notch

4- V-notch

6- Draw the fatigue curve (S/N) of a metal having the following surface characteristics: polished surface; sand-blasted surface; U-notches surface; V-notched surface. Explain differences.



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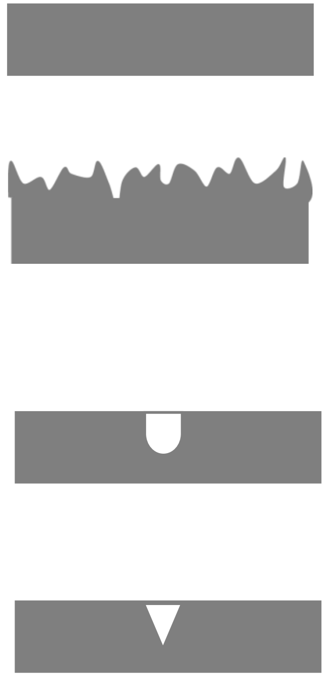

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